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The Ginzburg-Landau model

2.1 Introduction

The superconducting state and the normal metallic state are separate thermodynamic phases of matter in just the same way as gas, liquid and solid are different phases. Similarly, the normal Bose gas and BEC, or normal liquid He^4 and superfluid He II are separated by a thermodynamic phase transitions. Each such phase transition can be characterized by the nature of the singularities in specific heat and other thermodynamic variables at the transition, T_c . We can therefore examine the problems of superfluidity and superconductivity from the point of view of the thermodynamics of phase transitions.

The theory of superconductivity introduced by Ginzburg and Landau in 1950 describes the superconducting phase transition from this thermodynamic point of view. It was originally introduced as a phenomenological theory, but later Gor'kov showed that it can be derived from full the microscopic BCS theory in a suitable limit.

In this chapter we shall first discuss the superconducting phase transition from the point of view of equilibrium thermodynamics. Then we gradually build up towards the full Ginzburg Landau model. First we discuss spatially uniform systems, then spatially varying systems and finally systems in an external magnetic field. The Ginzburg Landau theory makes many useful and important predictions. Here we focus on just two applications: to flux quantization, and to the Abrikosov flux lattice in type II superconductors.

The Ginzubrg Landau theory as originally applied to superconductors was *par excellance* a **mean-field theory** of the thermodynamic state. However, in fact, one of its most powerful features is that it can be used to go beyond the original mean-field limit, so as to include the effects of thermal fluctuations. We shall see below that such fluctuations are largely negligible in the case of conventional “low- T_c ” superconductors, making the

¹In fact, the Ginzburg Landau model is very general and has applications in many different areas of physics. It can be modified to describe many different physical systems, including magnetism, liquid crystal phases and even the symmetry breaking phase transitions which took place in the early universe as matter cooled following the big bang!

mean-field approximation essentially exact. However in the newer high T_c superconductors these fluctuations lead to many important phenomena, such as flux flow, and the melting of the Abrikosov vortex lattice.

2.2 The condensation energy

We already have enough information about superconductivity to derive some important thermodynamic properties about the superconducting phase transition. We can analyze the phase diagram of superconductors in exactly the same manner as one would consider the well known thermodynamics of a liquid gas phase transition problem, such as given by the van der Waals equation of state. However, for the superconductor instead of the pair of thermodynamic variables P, V (pressure and volume) we have the magnetic variables $\mathbf{H}$ and $\mathbf{M}$ as the relevant thermodynamic parameters.

Let us first briefly review the basic thermodynamics of magnetic materials. This is covered in several undergraduate text books on thermodynamics such as Mandl (1987) or Callen (1960), Blundell (2001). If we consider a long cylindrical sample in a solenoidal field, as shown in Fig. 1.6, then the magnetic field $\mathbf{H}$ inside the sample is given by

$$\mathbf{H} = \frac{N}{L} I \mathbf{e}_z, \quad (2.1)$$

where the coil has N/L turns per metre, I is the current and $\mathbf{e}_z$ is a unit vector along the axis of the cylinder. The total work done, dW , on increasing the current infinitesimally from I to $I + dI$ can be calculated as

$$\begin{aligned} dW &= -N \int \mathcal{E} I dt \\ &= +N \int \frac{d\Phi}{dt} I dt \\ &= +N I d\Phi \\ &= +N A I dB \\ &= +N V \mathbf{H} \cdot d\mathbf{B} \\ &= +\mu_0 V (\mathbf{H} \cdot d\mathbf{M} + \mathbf{H} \cdot d\mathbf{H}) \end{aligned} \quad (2.2)$$

where A is the cross sectional area of the coil, $V = AL$ is its volume, $\mathcal{E} = -d\Phi/dt$ is the e.m.f. induced in the coil by the change in the total magnetic flux, Φ , through the sample. We also used the identity $\mathbf{B} = \mu_0(\mathbf{M} + \mathbf{H})$ in writing the last step in Eq. 2.2.

This analysis shows that we can divide the total work done by increasing the current in the coil into two separate parts. The first part,

$$\mu_0 \mathbf{H} \cdot d\mathbf{M}$$

per unit volume, is the **magnetic work** done on the sample. The second part,

$$\mu_0 \mathbf{H} \cdot d\mathbf{H}$$

is the work per unit volume which would have been done even if no sample had been present inside the coil; it is the work done by the **self-inductance** of the coil. If the coil is empty, $\mathbf{M} = 0$ and so $\mathbf{B} = \mu_0 \mathbf{H}$ and one can easily see that the work done is exactly the change in the vacuum field energy of the electromagnetic field

$$E_B = \frac{1}{2\mu_0} \int B^2 d^3r \quad (2.3)$$

due to the change of current in the solenoid coils. By convention² we shall not include this vacuum field energy, as work done “on the sample”. Therefore we define the magnetic work done on the sample as $\mu_0 \mathbf{H} \cdot d\mathbf{M}$ per unit volume.

With this definition of magnetic work the first law of thermodynamics for a magnetic material reads,

$$dU = TdS + \mu_0 V \mathbf{H} \cdot d\mathbf{M} \quad (2.4)$$

where U is the total internal energy, TdS is the heat energy with T the temperature and S the entropy. We see that the magnetic work is analogous to the work, $-PdV$, in a gas. As in the usual thermodynamics of gases the internal energy, U , is most naturally thought of as a function of the entropy and volume: $U(S, V)$. The analogue of the first law for a magnetic system, Eq. 2.4, shows that the internal energy of a magnetic substance is most naturally thought of as a function of S and $\mathbf{M}$, $U(S, \mathbf{M})$. In terms of this function the temperature and field $\mathbf{H}$ are given by

$$T = \frac{\partial U}{\partial S} \quad (2.5)$$

$$\mathbf{H} = \frac{1}{\mu_0 V} \frac{\partial U}{\partial \mathbf{M}}. \quad (2.6)$$

However S and $\mathbf{M}$ are usually not the most convenient variables to work with. In a solenoidal geometry such as Fig. 1.6 it is the H -field which is directly fixed by the current, not $\mathbf{M}$. It is therefore useful to define magnetic analogues of the Helmholtz and Gibbs free energies

$$F(T, \mathbf{M}) = U - TS \quad (2.7)$$

$$G(T, \mathbf{H}) = U - TS - \mu_0 V \mathbf{H} \cdot \mathbf{M}. \quad (2.8)$$

As indicated, the Gibbs free energy G is naturally viewed as a function of T and $\mathbf{H}$ since,

²Unfortunately there is no single standard convention used by all books and papers in this field. Different contributions to the total energy are either included or not, and so one must be very careful when comparing similar looking equations from different texts and research papers. Our convention follows Mandl (1987) and Callen (1960).

$$dG = -SdT - \mu_0 V \mathbf{M} \cdot d\mathbf{H}. \quad (2.9)$$

In terms of G one can calculate the entropy and magnetization,

$$S = -\frac{\partial G}{\partial T} \quad (2.10)$$

$$\mathbf{M} = -\frac{1}{\mu_0 V} \frac{\partial G}{\partial \mathbf{H}}. \quad (2.11)$$

$G(T, \mathbf{H})$ is usually the most convenient thermodynamic quantity to work with since T and $\mathbf{H}$ are the variables which are most naturally controlled experimentally. Furthermore from $G(T, \mathbf{H})$ one can also reconstruct the free energy, $F = G + \mu_0 V \mathbf{H} \cdot \mathbf{M}$ or the internal energy $U = F + TS$.

The Gibbs free energy allows us to calculate the free energy difference between the superconducting state and the normal state. Consider the H , T phase diagram of a type I superconductor, as sketched above in Fig. 2.1. We can evaluate the change in Gibbs free energy in the superconducting state by integrating along the vertical line drawn. Along this line $dT = 0$, and so, clearly,

$$G_s(T, H_c) - G_s(T, 0) = \int_0^{H_c} dG = -\mu_0 V \int_0^{H_c} \mathbf{M} \cdot d\mathbf{H},$$

where the subscript s implies that $G(T, \mathbf{H})$ is in the superconducting state. But for a type I superconductor in the superconducting state we know from the Meissner-Ochsenfeld effect that $\mathbf{M} = -\mathbf{H}$ and thus,

$$G_s(T, H_c) - G_s(T, 0) = \mu_0 \frac{H_c^2}{2} V.$$

Now, at the critical field H_c in Fig. 2.1 the normal state and the superconducting state are in thermodynamic equilibrium. Equilibrium between phases implies that the two Gibbs free energies are equal:

$$G_s(T, H_c) = G_n(T, H_c).$$

Furthermore, in the normal state $M \approx 0$ (apart from the small normal metal paramagnetism or diamagnetism which we neglect). So if the normal metal state had persisted below H_c down to zero field, it would have had a Gibbs free energy of,

$$G_n(T, H_c) - G_n(T, 0) = \int_0^{H_c} dG = -\mu_0 V \int_0^{H_c} M dH \approx 0.$$

Putting these together we find the difference in Gibbs free energies of superconducting and normal states at zero field:

$$G_s(T, 0) - G_n(T, 0) = -\mu_0 V \frac{H_c^2}{2} \quad (2.12)$$

The Gibbs potential for the superconducting state is lower, so it is the stable state.

We can also write the above results in terms of the more familiar Helmholtz free energy. Using $F = G - \mu_0 \mathbf{H} \cdot \mathbf{M}$ and substituting $\mathbf{H} = \mathbf{M} = 0$ we can see that the difference in Helmholtz free energies $F(T, \mathbf{M})$ is the same as for the Gibbs potentials, and hence

$$F_s(T, 0) - F_n(T, 0) = -\mu_0 V \frac{H_c^2}{2}. \quad (2.13)$$

The quantity $\mu_0 H_c^2/2$ is the **condensation energy**. It is a measure of the gain in free energy per unit volume in the superconducting state compared to the normal state at the same temperature.

As an example let's consider niobium. Here $T_c = 9\text{K}$, and $H_c = 160\text{kA/m}$ ($B_c = \mu_0 H_c = 0.2\text{T}$). The condensation energy $\mu_0 H_c^2/2 = 16.5\text{kJ/m}^3$. Given that Nb has a bcc crystal structure with a 0.33nm lattice constant we can work out the volume per atom and find that the condensation energy is only around $2\mu\text{eV/atom}$! Such tiny energies were a mystery until the BCS theory, which shows that the condensation energy is of order $(k_B T_c)^2 g(E_F)$, where $g(E_F)$ is the density of states at the Fermi level. The energy is so small because $k_B T_c$ is many orders of magnitude smaller than the Fermi energy, ϵ_F .

The similar thermodynamic arguments can also be applied to calculate the condensation energy of type II superconductors. Again the magnetic work per unit volume is calculated as an integral along a countour, as shown in the right panel of Fig. 2.1,

$$G_s(T, H_{c2}) - G_s(T, 0) = \mu_0 V \int_0^{H_{c2}} \mathbf{M} \cdot d\mathbf{H} \quad (2.14)$$

The integral is simply the area under the curve of M as a function of H drawn in Fig. 1.8 (assuming that $\mathbf{M}$ and $\mathbf{H}$ have the same vector directions).

Defining the value of H_c for a type II superconducting from the value of the integral

$$\frac{1}{2} H_c^2 \equiv \int_0^{H_{c2}} \mathbf{M} \cdot d\mathbf{H} \quad (2.15)$$

we again can express the zero field condensation energy in terms of H_c ,

$$F_s(T, 0) - F_n(T, 0) = -\mu_0 V \frac{H_c^2}{2}. \quad (2.16)$$

Here H_c is called the **thermodynamic critical field**. Note that there is no phase transition at H_c in a type II superconductor. The only real transitions are at H_{c1} and H_{c2} , and H_c is merely a convenient measure of the condensation energy.

We can also calculate the entropy of the superconducting state using the same methods. A simple calculation (Exercise 4.1) shows that in a type

In a superconductor there is a finite change in entropy between the normal and superconducting states at H_c ,

$$S_s(T, H_c) - S_n(T, H_c) = -\mu_0 H_c \frac{dH_c}{dT}. \quad (2.17)$$

This shows that the phase transition is generally **first-order**, i.e. it has a finite latent heat. But, in zero external field, at the point $(T, H) = (T_c, 0)$ in Fig. 2.1, this entropy difference goes to zero, and so in this case the phase transition is **second-order**.

2.3 Ginzburg-Landau theory of the bulk phase transition

The Ginzburg-Landau theory of superconductivity is built upon a general approach to the theory of second order phase transitions which Landau had developed in the 1930's. Landau had noticed that typically second order phase transitions, such as the Curie temperature in a ferromagnet, involve some change in symmetry of the system. For example a magnet above the Curie temperature, T_c , has no magnetic moment. But below T_c a spontaneous magnetic moment develops. In principle could point in any one of a number of different directions, each with an equal energy, but the system spontaneously chooses one particular direction. In Landau's theory such phase transitions are characterized by an **order parameter** which is zero in the disordered state above T_c , but becomes non-zero below T_c . In the case of a magnet the magnetization, $\mathbf{M}(\mathbf{r})$, is a suitable order parameter.

For superconductivity Ginzburg and Landau (GL) postulated the existence of an order parameter denoted by ψ . This characterizes the superconducting state, in the same way as the magnetization does in a ferromagnet. The order parameter is assumed to be some (unspecified) physical quantity which characterizes the state of the system. In the normal metallic state above the critical temperature T_c of the superconductor it is zero. While in the superconducting state below T_c it is non-zero. Therefore it is assumed to obey:

$$\psi = \begin{cases} 0 & T > T_c \\ \psi(T) \neq 0 & T < T_c. \end{cases} \quad (2.18)$$

Ginzburg and Landau postulated that the order parameter ψ should be a complex number, thinking of it as a macroscopic wave function for the superconductor in analogy with superfluid ^4He . At the time of their original work the physical significance of this complex ψ in superconductors was not at all clear. But, as we shall see below, in the microscopic BCS theory of superconductivity there appears a parameter, Δ , which is also complex. Gor'kov was able to derive the Ginzburg-Landau theory from BCS theory, and show that ψ is essentially the same as Δ , except for some constant numerical factors. In fact, we can even identify $|\psi|^2$ as the density of BCS "Cooper pairs" present in the sample.

Ginzburg and Landau assumed that the free energy of the superconductor must depend smoothly on the parameter ψ . Since ψ is complex and the

free energy must be real, the energy can only depend on $|\psi|$. Furthermore, since ψ goes to zero at the critical temperature, T_c , we can Taylor expand the free energy in powers of $|\psi|$. For temperatures close to T_c only the first two terms in the expansion should be necessary, and so the free energy density ($f = F/V$) must be of the form:

$$f_s(T) = f_n(T) + a(T)|\psi|^2 + \frac{1}{2}b(T)|\psi|^4 + \dots \quad (2.19)$$

since $|\psi|$ is small. Here $f_s(T)$ and $f_n(T)$ are the superconducting state and normal state free energy densities, respectively. Clearly Eq. 2.19 is the only possible function which is real for any complex ψ near $\psi = 0$ and which is a differentiable function of ψ and ψ^* near to $\psi = 0$. The parameters $a(T)$ and $b(T)$ are, in general, temperature dependent phenomenological parameters of the theory. However it is assumed that they must be smooth functions of temperature. We must also assume that $b(T)$ is positive, since otherwise the free energy density would have no minimum, which would be unphysical (or we would have to extend the expansion to include higher powers such as $|\psi|^6$).

Plotting $f_s - f_n$ as a function of ψ is easy to see that there are two possible curves, depending on the sign of the parameter $a(T)$, as shown in Fig. 2.2. In the case $a(T) > 0$, the curve has one minimum at $\psi = 0$. On the other hand, for $a(T) < 0$ there are minima wherever $|\psi|^2 = -a(T)/b(T)$. Landau and Ginzburg assumed that at high temperatures, above T_c , we have $a(T)$ positive, and hence the minimum free energy solution is one with $\psi = 0$, i.e. the normal state. But if $a(T)$ gradually decreases as the temperature T is reduced, then the state of the system will change suddenly when we reach the point $a(T) = 0$. Below this temperature the minimum free energy solution changes to one with $\psi \neq 0$. Therefore we can identify the temperature where $a(T)$ becomes zero as the critical temperature T_c .

Near to this critical temperature, T_c , assuming that the coefficients $a(T)$ and $b(T)$ change smoothly with temperature, we can make a Taylor expansion,

$$\begin{aligned} a(T) &\approx \dot{a} \times (T - T_c) + \dots \\ b(T) &\approx b + \dots, \end{aligned} \quad (2.20)$$

where $\dot{a}$ and b are two phenomenological constants. Then for temperatures just above T_c , $a(T)$ will be positive, and we have the free energy minimum, $\psi = 0$. On the other hand, just below T_c we will have minimum energy solutions with non-zero $|\psi|$, as seen in Fig. 2.2. In terms of the parameters $\dot{a}$ and b it is easy to see that

$$|\psi| = \begin{cases} \left(\frac{\dot{a}}{b}\right)^{1/2} (T_c - T)^{1/2} & T < T_c \\ 0 & T > T_c \end{cases}. \quad (2.21)$$

The corresponding curve of $|\psi|$ as a function of temperature, T , is shown in Fig 2.3. One can see the abrupt change from zero to non-zero values at

the critical temperature T_c . In fact, this curve is qualitatively similar to those obtained with other types of second order phase transitions within Landau's general theory. For example the behaviour of the order parameter ψ near T_c in Fig. 2.3 resembles closely change in the magnetization $\mathbf{M}$ in a ferromagnet near its Curie point in the Stoner theory of magnetism (Blundell 2001).

It turns out to be very important that, because ψ is complex, there are in fact an infinite set of minima corresponding to all possible values of the complex phase θ ,

$$\psi = |\psi|e^{i\theta}. \quad (2.22)$$

The phase value, θ is arbitrary, since all values lead to the same total free energy. But, just as in the case of the direction of magnetization $\mathbf{M}$ in a ferromagnet the system spontaneously chooses one particular value. A magnet heated to above T_c and then cooled again will almost certainly adopt a different random direction of magnetization, and the same would be true for the angle θ in a superconductor. In fact we have met this same concept before, in Chapter 2, when we discussed the XY symmetry of the macroscopic wave function in superfluid He II (Fig. ??).

The value of the minimum free energy in Fig. 2.2, is easily found to be $-a(T)^2/2b(T)$. This is the free energy difference (per unit volume) between the superconducting and non-superconducting phases of the system at temperature T . This corresponds to the condensation energy of the superconductor, and so we can write

$$f_s(T) - f_n(T) = -\frac{\dot{a}^2(T - T_c)^2}{2b} = -\mu_0 \frac{H_c^2}{2}, \quad (2.23)$$

giving the thermodynamic critical field,

$$H_c = \frac{\dot{a}}{(\mu_0 b)^{1/2}}(T_c - T) \quad (2.24)$$

near to T_c .

From this free energy we can also obtain other relevant physical quantities, such as the entropy and heat capacity. Differentiating f with respect to T gives the entropy per unit volume, $s = S/V$,

$$s_s(T) - s_n(T) = -\frac{\dot{a}^2}{b}(T_c - T), \quad (2.25)$$

below T_c . At T_c there is no discontinuity in entropy, or latent heat, confirming that the Ginzburg-Landau model corresponds to a second order thermodynamic phase transition. But there is a sudden change in specific heat at T_c . Differentiating the entropy to find the heat capacity $C_V = Tds/dT$ per unit volume we obtain

$$C_{Vs} - C_{Vn} = \begin{cases} T \frac{\dot{a}^2}{b} & T < T_c \\ 0 & T > T_c \end{cases} \quad (2.26)$$

and so the heat capacity has a discontinuity

$$\Delta C_V = T_c \frac{\dot{a}^2}{b} \quad (2.27)$$

at T_c . The metallic normal state heat capacity is linear in T , $C_{Vn} = \gamma T$, with γ the Sommerfeld constant, and so the full heat capacity curve looks like Fig. 2.4 near to T_c .³

Interestingly the specific heat for superconductors shown in Fig. 2.4 is qualitatively quite very different from both the case of Bose-Einstein condensation, shown in Fig. ??, and the λ point of superfluid ⁴He, Fig. ??.⁴

2.4 Ginzburg-Landau theory of inhomogenous systems

The complete Ginzburg and Landau theory of superconductivity also allows for the possibility that the order parameter depends on position, $\psi(\mathbf{r})$. This, of course, now really begins to resemble the macroscopic condensate wave function introduced in chapter 2 for the case of superfluid helium.

Ginzburg and Landau postulated that the Free energy is as given above, together with a new term depending on the gradient of $\psi(\mathbf{r})$. With this term free energy density becomes,

$$f_s(T) = f_n(T) + \frac{\hbar^2}{2m^*} |\nabla \psi(\mathbf{r})|^2 + a(T) |\psi(\mathbf{r})|^2 + \frac{b(T)}{2} |\psi(\mathbf{r})|^4 \quad (2.28)$$

at point $\mathbf{r}$ in the absence of any magnetic fields. Setting $\psi(\mathbf{r})$ to a constant value, $\psi(\mathbf{r}) = \psi$, we see that the parameters $a(T)$ and $b(T)$ are the same as for the bulk theory described in the previous section. The new parameter m^* determines the energy cost associated with gradients in $\psi(\mathbf{r})$. It has dimensions of mass, and it plays the role of an effective mass for the quantum system with macroscopic wave function $\psi(\mathbf{r})$.

In order to find the order parameter $\psi(\mathbf{r})$ we must minimize the total free energy of the system,

$$F_s(T) = F_n(T) + \int d^3r \left(\frac{\hbar^2}{2m^*} |\nabla \psi|^2 + a(T) |\psi(\mathbf{r})|^2 + \frac{b(T)}{2} |\psi(\mathbf{r})|^4 \right) d^3r. \quad (2.29)$$

To find the minimum we must consider an infinitesimal variation in the function $\psi(\mathbf{r})$

³The Ginzburg Landau theory can only be reliably used at temperatures close to T_c . Therefore our calculated specific heat is only correct near to T_c , and we cannot legitimately continue the Ginzburg-Landau line in Fig. 2.4 down from T_c all the way to $T = 0$.

⁴In fact the differences are deceptive! Our theory rests on a mean-field approximation and has neglected important thermal fluctuation effects, as we shall see below. When these fluctuations are large, as in the case of high temperature superconductors, the observed specific heat near T_c appears to show exactly the same XY universality class as the lambda point in superfluid helium (Overend 1994).

$$\psi(\mathbf{r}) \rightarrow \psi(\mathbf{r}) + \delta\psi(\mathbf{r}) \quad (2.30)$$

relative to some function $\psi(\mathbf{r})$. Evaluating the change in the total free energy due to $\delta\psi$ and dropping all terms of higher than linear order in the variation $\delta\psi$ we find after some lengthy algebra

$$\begin{aligned} \delta F_s = & \int \left[\frac{\hbar^2}{2m^*} (\nabla \delta\psi^*) \cdot (\nabla \psi) + \delta\psi^* (a\psi + b\psi|\psi^2|) \right] d^3r \\ & + \int \left[\frac{\hbar^2}{2m^*} (\nabla \psi^*) \cdot (\nabla \delta\psi) + (a\psi^* + b\psi^*|\psi^2|) \delta\psi \right] d^3r. \end{aligned} \quad (2.31)$$

The two terms involving gradients can be integrated by parts, to obtain

$$\begin{aligned} \delta F_s = & \int \delta\psi^* \left(-\frac{\hbar^2}{2m^*} \nabla^2 \psi + a\psi + b\psi|\psi^2| \right) d^3r \\ & + \int \left(-\frac{\hbar^2}{2m^*} \nabla^2 \psi + a\psi + b\psi|\psi^2| \right)^* \delta\psi d^3r. \end{aligned} \quad (2.32)$$

The condition for $\psi(\mathbf{r})$ to produce a minimum in free energy is that $\delta F = 0$ for any arbitrary variation $\delta\psi(\mathbf{r})$. From Eq. 2.32 this can only be when $\psi(\mathbf{r})$ obeys

$$-\frac{\hbar^2}{2m^*} \nabla^2 \psi + a\psi + b\psi|\psi^2| = 0. \quad (2.33)$$

We can obtain this same result more formally by noting that the total Free energy of the solid is a **functional** of $\psi(\mathbf{r})$, denoted $F_s[\psi]$, meaning that the scalar number F_s depends on the whole function $\psi(\mathbf{r})$ at all points in the system, $\mathbf{r}$. It will be minimized by a function $\psi(\mathbf{r})$ which satisfies

$$\frac{\partial F_s[\psi]}{\partial \psi(\mathbf{r})} = 0 \quad \frac{\partial F_s[\psi]}{\partial \psi^*(\mathbf{r})} = 0. \quad (2.34)$$

where the derivatives are mathematically **functional derivatives**. Functional derivative can be defined by analogy with the idea of a partial derivative. For a function of many variables, $f(x_1, x_2, x_3, \dots)$ we can express changes in the function value due to infinitesimal variations of the parameters using the standard expression

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \frac{\partial f}{\partial x_3} dx_3 + \dots \quad (2.35)$$

Considering the free energy as a function of infinitely many variables, $\psi(\mathbf{r})$ and $\psi^*(\mathbf{r})$ at all possible points $\mathbf{r}$ we can write the analogue of Eq. 2.35 as,

$$dF_s = \int \left(\frac{\partial F_s[\psi]}{\partial \psi(\mathbf{r})} d\psi(\mathbf{r}) + \frac{\partial F_s[\psi]}{\partial \psi^*(\mathbf{r})} d\psi^*(\mathbf{r}) \right) d^3r. \quad (2.36)$$

In comparison with Eq. 2.32 we see that

$$\frac{\partial F_s[\psi]}{\partial \psi^*(\mathbf{r})} = -\frac{\hbar^2}{2m^*} \nabla^2 \psi + a(T)\psi + b(T)\psi|\psi^2| \quad (2.37)$$

and

$$\frac{\partial F_s[\psi]}{\partial \psi(\mathbf{r})} = \left(-\frac{\hbar^2}{2m^*} \nabla^2 \psi + a(T)\psi + b(T)\psi|\psi^2|\right)^* \quad (2.38)$$

which is just the complex conjugate of Eq. 2.37. Perhaps it seems surprising that we can effectively treat $\psi(\mathbf{r})$ and $\psi^*(\mathbf{r})$ as independent variables in the differentiation, but this is correct because there are two independent real functions, $\text{Re}[\psi(\mathbf{r})]$ and $\text{Im}[\psi(\mathbf{r})]$, which can be varied separately.

Thus we have found that minimizing the total Free energy leads to the following Schrödinger like equation for $\psi(\mathbf{r})$,

$$-\frac{\hbar^2}{2m^*} \nabla^2 \psi(\mathbf{r}) + (a + b|\psi(\mathbf{r})|^2) \psi(\mathbf{r}) = 0. \quad (2.39)$$

However, unlike the usual Schrödinger equation, this is a non-linear equation because of the second term in the bracket. Because of this non-linearity the quantum mechanical principle of superposition does not apply, and the normalization of ψ is different from the usual one in quantum mechanics.

2.5 Surfaces of Superconductors

The effective non-linear Schrödinger equation, Eq. 2.39, has several useful applications. In particular, it can be used to study the response of the superconducting order parameter to external perturbations. Important examples of this include the properties of the surfaces and interfaces of superconductors.

Consider a simple model for the interface between a normal metal and a superconductor. Suppose that the interface lies in the yz plane separating the normal metal in the $x < 0$ region from the superconductor in the $x > 0$ region. On the normal metal side of the interface the superconducting order parameter, $\psi(\mathbf{r})$, must be zero. Assuming that $\psi(\mathbf{r})$ must be continuous, we must therefore solve the non-linear Schrödinger equation,

$$-\frac{\hbar^2}{2m^*} \frac{d^2 \psi(x)}{dx^2} + a(T)\psi(x) + b(T)\psi^3(x) = 0 \quad (2.40)$$

in the region $x > 0$ with the boundary condition at $\psi(0) = 0$. It turns out that one can solve this equation directly (exercise 4.2) to find

$$\psi(x) = \psi_0 \tanh\left(\frac{x}{\sqrt{2}\xi(T)}\right), \quad (2.41)$$

as shown in Fig. 2.5. Here ψ_0 is the value of the order parameter in the bulk far from the surface and the parameter $\xi(T)$ is defined by

$$\xi(T) = \left(\frac{\hbar^2}{2m^*|a(T)|} \right)^{1/2}. \quad (2.42)$$

This quantity, which has dimensions of length, is called the **Ginzburg-Landau coherence length**. It is an important physical parameter characterizing the superconductor. In Fig. 2.5 one can see that $\xi(T)$ is a measure of the distance from the surface over which the order parameter has recovered back to nearly its bulk value.

The Ginzburg-Landau coherence length arises in almost all problems of inhomogeneous superconductors, including surfaces, interfaces, defects and vortices. Using $a(T) = \dot{a}(T - T_c)$ the coherence length $\xi(T)$ can be rewritten,

$$\xi(T) = \xi(0)|t|^{-1/2}, \quad (2.43)$$

where

$$t = \frac{T - T_c}{T_c} \quad (2.44)$$

is called the reduced temperature. This expression makes it clear that the coherence length $\xi(T)$ diverges at the critical temperature T_c , and that its divergence is characterized by a critical exponent of $1/2$. This exponent is typical for mean-field theories such as the Ginzburg-Landau model. The zero temperature value of ξ , $\xi(0)$, is apart from some numerical factors of order unity, essentially the same as the Pippard coherence length for superconductors, as introduced in Chapter 3. In BCS theory the coherence length relates to the physical size of a single Cooper pair.

It is also possible to calculate the contribution to the total free energy due to the surface in Fig. 2.5. The surface contribution to the total free energy is

$$\sigma = \int_0^\infty \left(\frac{\hbar^2}{2m^*} \left(\frac{d\psi(x)}{dx} \right)^2 + a\psi^2(x) + \frac{b}{2}\psi^4(x) + \frac{1}{2}\mu_0 H_c^2 \right) dx \quad (2.45)$$

with $\psi(x)$ given by Eq. 2.41. Here $-\mu_0 H_c^2/2 = -a^2/2b$ is the bulk free energy density. Evaluating the integral (de Gennes 1960) gives

$$\sigma = \frac{1}{2}\mu_0 H_c^2 \times 1.89\xi(T) \quad (2.46)$$

free energy per unit area of the surface.

This theory can also be used to model the **proximity effect** between two superconductors. At an interface between two different superconducting materials the one with the higher T_c will become superconducting first, and will nucleate superconductivity at the surface of the second one. Superconductivity will nucleate at temperatures above the T_c for the second superconductor. If one makes the lower T_c superconductor a thin layer, of order the coherence length $\xi(T)$ in thickness, then the whole system will

become superconducting at a temperature above the natural critical temperature of the lower T_c material. Effectively the order parameter $\psi(\mathbf{r})$ has been forced to become non-zero in the thin film material by its proximity to the higher T_c material.

2.6 Ginzburg-Landau theory in a magnetic field

The full power of the GL approach to superconductors only becomes apparent when we include one final term, the effect of a magnetic field. It is this which truly shows that this is a fully fledged theory of superconductivity, complete with Meissner-Ochsenfeld effect, London equation and so on. Effectively, the Ginzburg-Landau theory as developed given in the previous sections did not include any effects of the charge of the superconducting condensate. Therefore it would be appropriate for systems of neutral particles, such as a superfluid, or for situations where there are no supercurrents. But in the presence of supercurrents of charged particles we must extend the theory to include the interaction of the current and magnetic field.

What is needed is to include the effects of magnetic fields in the free energy. Ginzburg and Landau postulated that the magnetic field enters as if $\psi(\mathbf{r})$ were the wave function for charged particles, i.e with the usual replacement in quantum mechanics

$$\frac{\hbar}{i}\nabla \rightarrow \frac{\hbar}{i}\nabla - q\mathbf{A} \quad (2.47)$$

where q is the charge and $\mathbf{A}$ is the magnetic vector potential. For all known superconductors it turns out that the appropriate charge q is $-2e$. Why this is the case only became clear after the BCS theory was developed, and the link between the BCS theory and the Ginzburg-Landau model had been explained by Gor'kov. He showed that the correct physical interpretation of the GL order parameter $\psi(\mathbf{r})$ is that it can be understood as the wave function for the centre of mass motion of Cooper pairs of electrons. Since each Cooper pair has a net charge of $-2e$, then this is the correct effective charge q .⁵ Actually the sign can equally well be taken as $q = +2e$, since we can think of Cooper pairs of holes as readily as pairs of electrons. In fact no observable effects in the Ginzburg-Landau theory differ when we take a different convention for the sign.

With this replacement the Ginzburg-Landau free energy density of the superconductor becomes,

$$f_s(T) = f_n(T) + \frac{\hbar^2}{2m^*} |(\frac{\hbar}{i}\nabla + 2e\mathbf{A})\psi|^2 + a|\psi|^2 + \frac{b}{2}|\psi|^4. \quad (2.48)$$

⁵In fact, in their original paper Ginzburg and Landau assumed that the effective charge would be e , not $2e$. Reputedly, Ginzburg, then a young PhD student, told his famous advisor Landau that $2e$ fitted the available experimental data better than e , but Landau overruled him and insisted that he was sure that it must be e !

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To obtain the total free energy we must integrate this over the system, but we must also include an additional term, corresponding to the electromagnetic field energy of the field $\mathbf{B}(\mathbf{r}) = \nabla \times \mathbf{A}$ at each point $\mathbf{r}$. Therefore the total free energy of both the superconductor and the magnetic field is,⁶

$$F_s(T) = F_n(T) + \int \left(\frac{\hbar^2}{2m^*} \left| \left(\frac{\hbar}{i} \nabla + 2e\mathbf{A} \right) \psi \right|^2 + a|\psi|^2 + \frac{b}{2}|\psi|^4 \right) d^3r + \frac{1}{2\mu_0} \int B(\mathbf{r})^2 d^3r. \quad (2.49)$$

The first integral is carried out over points $\mathbf{r}$ inside the sample, while the second is performed over all space.

The condition for the minimum free energy state is again found by performing a functional differentiation to minimize with respect to $\psi(\mathbf{r})$ and $\psi^*(\mathbf{r})$. The resulting equation for $\psi(\mathbf{r})$ is again a non-linear Schrödinger equation, but now with a term containing the magnetic vector potential $\mathbf{A}$,

$$-\frac{\hbar^2}{2m^*} (\nabla + \frac{2ei}{\hbar} \mathbf{A})^2 \psi(\mathbf{r}) + (a + b|\psi|^2) \psi(\mathbf{r}) = 0. \quad (2.50)$$

The supercurrents due to the magnetic field can be found from functional derivative of the Ginzburg Landau superconductor free energy with respect to the vector potential,

$$\mathbf{j}_s = - \frac{\partial F_s}{\partial \mathbf{A}(\mathbf{r})} \quad (2.51)$$

which leads to the supercurrent

$$\mathbf{j}_s = - \frac{2e\hbar i}{2m^*} (\psi^* \nabla \psi - \psi \nabla \psi^*) - \frac{(2e)^2}{m^*} |\psi|^2 \mathbf{A}. \quad (2.52)$$

Note the close similarity to the superfluid current flow that we found earlier in the case of ^4He , Eq. ???. The differences from Eq. ??? are firstly the charge of the condensate particles, $-2e$, and the presence of the last term which provides the effect of the vector potential $\mathbf{A}$. Finally, the vector potential must be obtained from the magnetic field arising from both the supercurrents and any other currents, such as the external currents, $\mathbf{j}_{ext}$, in the solenoid coils of Fig. 1.6,

$$\nabla \times \mathbf{B} = \mu_0(\mathbf{j}_{ext} + \mathbf{j}_s), \quad (2.53)$$

as given by Maxwell's equations.

⁶Note that in the definition of the magnetic work given in Eq. 2.4, we excluded the part of this field energy, $\mu_0 H^2/2$ per unit volume, that would be present even with no sample present inside the coil in Fig. 1.6. From now on, it will be more convenient to include this energy explicitly, making F_s the total free energy of both sample and vacuum fields.

2.7 Gauge Symmetry and Symmetry Breaking

The Ginzburg-Landau order parameter for superconductors has both an amplitude and a complex phase

$$\psi(\mathbf{r}) = |\psi(\mathbf{r})|e^{i\theta(\mathbf{r})}. \quad (2.54)$$

This is similar to the macroscopic wave function for superfluid He II, introduced in Chapter 2. However, unlike superfluids of neutral particles, something very interesting happens now when we consider gauge invariance.

If we make a **gauge transformation** of the magnetic vector potential

$$\mathbf{A}(\mathbf{r}) \rightarrow \mathbf{A}(\mathbf{r}) + \nabla\chi(\mathbf{r}) \quad (2.55)$$

then we must make a corresponding change in the phase of the order parameter, θ . Consider the term in the Ginzburg-Landau free energy density containing the canonical momentum operator

$$\hat{p} = \frac{\hbar}{i}\nabla + 2e\mathbf{A}.$$

If we change the phase of the order parameter by

$$\psi(\mathbf{r}) \rightarrow \psi(\mathbf{r})e^{i\theta(\mathbf{r})} \quad (2.56)$$

then we obtain

$$\begin{aligned} \hat{p}\psi(\mathbf{r})e^{i\theta(\mathbf{r})} &= e^{i\theta(\mathbf{r})}\left(\frac{\hbar}{i}\nabla + 2e\mathbf{A}\right)\psi(\mathbf{r}) + \psi(\mathbf{r})e^{i\theta(\mathbf{r})}\hbar\nabla\theta(\mathbf{r}) \\ &= e^{i\theta(\mathbf{r})}\left(\frac{\hbar}{i}\nabla + 2e\left(\mathbf{A} + \frac{\hbar}{2e}\nabla\theta\right)\right)\psi(\mathbf{r}). \end{aligned} \quad (2.57)$$

From this it follows that the free energy will be unchanged when we simultaneously change $\psi(\mathbf{r})$ to $\psi(\mathbf{r})e^{i\theta(\mathbf{r})}$ and the vector potential according to,

$$\mathbf{A}(\mathbf{r}) \rightarrow \mathbf{A}(\mathbf{r}) + \frac{\hbar}{2e}\nabla\theta. \quad (2.58)$$

This shows that the theory satisfies **local gauge invariance**. Both the phase of the order parameter and the magnetic vector potential depend on the choice of gauge, but all physical observables (free energy, magnetic field $\mathbf{B}$ etc.) are gauge invariant.

So far this is all completely general. But we saw earlier that a bulk superconductor has a ground state with a constant order parameter, ψ . Therefore it must have the same θ everywhere. There must be a **phase-stiffness**, or an energy cost associated with changing θ from one part of the solid to another. If we consider a superconductor in which the order parameter has a constant magnitude, $|\psi|$, and a phase $\theta(\mathbf{r})$ which varies

only slowly with position $\mathbf{r}$, then (using Eq. 2.57) we obtain the total free energy

$$F_s = F_s^0 + \rho_s \int d^3r \left(\nabla\theta + \frac{2e}{\hbar} \mathbf{A} \right)^2. \quad (2.59)$$

Here the **superfluid stiffness** is defined by,

$$\rho_s = \frac{\hbar^2}{2m^*} |\psi|^2 \quad (2.60)$$

and F_s^0 is the total free energy in the ground state ($\theta = \text{constant}$, $\mathbf{A} = 0$). Now if we choose some particular gauge for $\mathbf{A}(\mathbf{r})$, such as the London gauge, $\nabla \cdot \mathbf{A} = 0$, then within this fixed gauge there is now a free energy cost associated with further gradients in $\theta(\mathbf{r})$. To minimize the gradient energy, we must minimize the gradients, by making $\theta(\mathbf{r})$ as constant as possible throughout the system. In the case of zero applied magnetic field, we can choose $\mathbf{A} = 0$, and clearly then $\theta(\mathbf{r})$ will be constant everywhere in the system. Again we are back to the XY symmetry of Fig. ???. Since the system effectively chooses an (arbitrary) constant order parameter phase everywhere in the system, we can say that the system exhibits **long ranged order** in the order parameter phase, just as a ferromagnet has long ranged order in its magnetization $\mathbf{M}(\mathbf{r})$.

Because the long ranged order is in the phase variable (which is not normally a physical observable in quantum mechanics) we say that the system has **spontaneously broken global gauge symmetry**. The point is that *global* gauge symmetry refers to changing $\theta(\mathbf{r})$ by a constant amount everywhere in the whole solid (which does not require any change in $\mathbf{A}$). This is in contrast to *local* gauge symmetry in which $\theta(\mathbf{r})$ and $\mathbf{A}(\mathbf{r})$ are changed simultaneously, consistent with Eq. 2.58.

Eq. 2.59 also implies the London equation, and hence the Meissner-Ochsenfeld effect, bringing us full circle back to Chapter 3. The current can be calculated from a functional derivative of the free energy

$$\begin{aligned} \mathbf{j}_s &= -\frac{\partial F_s[\mathbf{A}]}{\partial \mathbf{A}(\mathbf{r})} \\ &= -\frac{2e}{\hbar} \rho_s \left(\nabla\theta + \frac{2e}{\hbar} \mathbf{A} \right). \end{aligned} \quad (2.61)$$

Starting in the ground state, where θ is constant, we directly find that with a small constant external vector potential, $\mathbf{A}$, the current is,

$$\mathbf{j}_s = -\rho_s \frac{(2e)^2}{\hbar^2} \mathbf{A} \quad (2.62)$$

which is exactly the same as the London equation. The superfluid stiffness, ρ_s , is essentially just the London superfluid density, n_s , in disguise!

To make the connection between ρ_s and the London superfluid fraction, n_s , more clear, consider the London equation

$$\mathbf{j}_s = -\frac{n_s e^2}{m_e} \mathbf{A}. \quad (2.63)$$

If we rewrite Eq. 2.61 in the form

$$\mathbf{j}_s = -\frac{(2e)^2}{2m^*} |\psi|^2 \mathbf{A}, \quad (2.64)$$

then clearly these are the same. It is conventional to define the constants so that the London superfluid density is $n_s = 2|\psi|^2$ and the Ginzburg-Landau effective mass is $m^* = 2m_e$ (where m_e is the bare electron mass). With this choice the equation can be interpreted physically as implying that $|\psi|^2$ is the density of pairs of electrons in the ground state. Therefore in comparison with the BCS theory of superconductivity we can interpret $|\psi|^2$ with the density of Cooper pairs in the ground state, and n_s as the density of electrons belonging to these Cooper pairs. The normal fraction, $n_n = n - n_s$ corresponds to the density of unpaired electrons. The Ginzburg-Landau parameter m^* is the mass of the Cooper pair, which is naturally twice the original electron mass.

In terms of the original free energy Ginzburg-Landau parameters, $\dot{a}$ and b , the superfluid density, n_s is given by

$$n_s = 2|\psi|^2 = 2 \frac{\dot{a}(T_c - T)}{b}. \quad (2.65)$$

Therefore the London penetration depth, $\lambda(T)$ is given by

$$\lambda(T) = \left(\frac{m_e b}{2\mu_0 e^2 \dot{a}(T_c - T)} \right)^{1/2}. \quad (2.66)$$

Clearly this will diverge at the critical temperature, T_c , since it is proportional to $(T_c - T)^{-1/2}$. We saw earlier that the Ginzburg Landau coherence length, $\xi(T)$, also diverges with the same power of $(T_c - T)$, and so the dimensionless ratio,

$$\kappa = \frac{\lambda(T)}{\xi(T)}, \quad (2.67)$$

is independent of temperature within the Ginzburg Landau theory. Table 2.1 summarizes the measured values of penetration depth and coherence length at zero temperature, $\lambda(0)$, $\xi(0)$, for a selection of superconductors.

2.8 Flux quantization

Let us now apply the Ginzburg-Landau theory to the case of a superconducting ring, as shown in Fig. 1.4. Describing the system using cylindrical polar coordinates, $\mathbf{r} = (r, \phi, z)$, with the z -axis perpendicular to the plane

Table 2.1 Penetration depth, $\lambda(0)$, and coherence length, $\xi(0)$, at zero temperature for some important superconductors. Data values are taken from Poole (2000).

	T_c (K)	$\lambda(0)$ (nm)	$\xi(0)$ (nm)	κ
Al	1.18	1550	45	0.03
Sn	3.72	180	42	0.23
Pb	7.20	87	39	0.48
Nb	9.25	39	52	1.3
Nb ₃ Ge	23.2	3	90	30
YNi ₂ B ₂ C	15	8.1	103	12.7
K ₃ C ₆₀	19.4	2.8	240	95
YBa ₂ Cu ₃ O _{7-δ}	91	1.65	156	95

of the ring, we see that the order parameter $\psi(\mathbf{r})$ must be periodic in the angle ϕ ,

$$\psi(r, \phi, z) = \psi(r, \phi + 2\pi, z). \quad (2.68)$$

We assume that the variations of $\psi(\mathbf{r})$ across the cross section of the ring are unimportant, and so we can neglect r and z dependence. Therefore the possible order parameters inside the superconductor are of the form

$$\psi(\phi) = \psi_0 e^{in\phi} \quad (2.69)$$

where n is an integer and ψ_0 is a constant. We can interpret n as a **winding number** of the macroscopic wave function, exactly as for the case of superfluid helium in Fig. ??.

However, unlike the case of superfluid helium, a circulating current in a superconductor will induce magnetic fields. Assuming that there is a magnetic flux Φ through the ring, then the vector potential can be chosen to be in the tangential direction, $\mathbf{e}_\phi$ and is given by

$$A_\phi = \frac{\Phi}{2\pi R}, \quad (2.70)$$

where R is the radius of the area enclosed by the ring. This follows from

$$\Phi \equiv \int \mathbf{B} \cdot d\mathbf{S} = \int (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint \mathbf{A} \cdot d\mathbf{r} = 2\pi R A_\phi. \quad (2.71)$$

The free energy corresponding to this wave function and vector potential is

$$F_s(T) = F_n(T) + \int d^3r \left(\frac{\hbar^2}{2m^*} |(\nabla + \frac{2ei}{\hbar} \mathbf{A})\psi|^2 + a|\psi|^2 + \frac{b}{2}|\psi|^4 \right) + E_B$$

$$= F_s^0(T) + V \left(\frac{\hbar^2}{2m^*} \left| \frac{in}{R} - \frac{2ei\Phi}{2\pi\hbar R} \right|^2 |\psi|^2 \right) + \frac{1}{2\mu_0} \int B^2 d^3r \quad (2.72)$$

where we have used the expression to gradient in cylindrical polar coordinates

$$\nabla X = \frac{\partial X}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial X}{\partial \phi} \mathbf{e}_\phi + \frac{\partial X}{\partial z} \mathbf{e}_z \quad (2.73)$$

(Boas 1983), V is the total volume of the superconducting ring, and $F_s^0(T)$ is the ground state free energy of the ring in the absence of any currents and magnetic fluxes. The vacuum magnetic field energy $E_B = (1/2\mu_0) \int B^2 d^3r$ can be expressed in terms of the inductance, L of the ring and the current I ,

$$E_B = \frac{1}{2} LI^2. \quad (2.74)$$

Clearly, it will be proportional to the square of the total flux, Φ through the ring

$$E_B \propto \Phi^2.$$

On the other hand, the energy of the superconductor contains a term depending on both the flux ϕ and the winding number, n . This term can be expressed as,

$$V \frac{\hbar^2}{2m^* R^2} |\psi|^2 (\Phi - n\Phi_0)^2,$$

where the **flux quantum** is $\Phi_0 = h/2e = 2.07 \times 10^{-15} \text{Wb}$.

We therefore see that the free energy is equal to the bulk free energy plus two additional terms depending only on the winding number n and the flux Φ . The energy of the superconducting ring is therefore of the general form,

$$F_s(T) = F_s^{\text{bulk}}(T) + \text{const.}(\Phi - n\Phi_0)^2 + \text{const.}\Phi^2. \quad (2.75)$$

This energy is sketched in Fig. 2.6. We can see from the figure that the free energy is a minimum whenever the flux through the loop obeys $\Phi = n\Phi_0$. This is the phenomenon of **flux quantization** in superconductors.

Taking a ring in its normal state above T_c and cooling it to below T_c will result in the system adopting one of the meta-stable minima in Fig. 2.6, depending on the applied field. It will then be trapped in the minimum, and a persistent current will flow around the ring to maintain a constant flux $\Phi = n\Phi_0$. Even if any external magnetic fields are turned off, the current in the ring must maintain a constant flux Φ in the ring. It is possible to directly measure the magnetic flux directly in such rings, and hence confirm that it is indeed quantized in units of Φ_0 , or multiples of $2 \times 10^{-15} \text{Wb}$. Incidentally, the fact that flux quantization is observed in units of $\Phi_0 = h/2e$ and not units of h/e is clear experimental proof that the relevant charge is $2e$ not e , hence implying the existence of Cooper pairs.

Given that a system is prepared in one of the meta-stable minima, it can, in principle, escape over the energy barriers to move into a neighbouring lower energy minimum. This would be a mechanism for the persistent current to decay, and hence for dissipation. Such an event corresponds to a change in the winding number, n , and is called a **phase-slip**. However the rate for thermally hopping over these barriers is exponentially small, of order

$$\frac{1}{\tau} \sim e^{-E_0/k_B T} \quad (2.76)$$

where E_0 is the barrier height between minima in Fig. 2.6. Clearly this thermal hopping rate can be made negligibly small. For example, E_0 is formally proportional to the ring volume, V , and so can be made arbitrarily large in a macroscopic system. In practice persistent currents have been observed to flow for years, with essentially no decay!

Another interesting possible mechanism for a phase slip would be a quantum tunnelling from one minimum to another. This would be possible at any temperature. But again the rate is impractically small in macroscopic systems. However, a very interesting recent development has been the direct observation of these tunnelling events in small **mesoscopic** superconducting rings. These experiments have demonstrated **macroscopic quantum coherence** and are discussed briefly in the next chapter.

2.9 The Abrikosov flux lattice

The great beauty of the Ginzburg-Landau theory is that it allows one to solve many difficult problems in superconductivity, without any reference to the underlying microscopic BCS theory. In some sense one could argue that it is more general, for example it would almost certainly apply to exotic superconductors, such as the high T_c cuprates, even though the original BCS theory does not seem to explain these systems. The other great advantage of the Ginzburg-Landau theory is that it is considerably easier to work with than the BCS theory, especially in cases where the order parameter has complicated spatial variations. The *tour de force* example of this is the **Abrikosov flux lattice**.

Abrikosov found a solution to the Ginzburg-Landau equations in the case of a bulk superconductor in a magnetic field. The result he obtained is remarkable in many respects. It is essentially an exact solution for type II superconductors, valid close to H_{c2} . Furthermore, it predicted a striking result, that just below H_{c2} the order parameter forms into a periodic structure of vortices. Each vortex carries a magnetic flux, hence explaining how magnetic flux enters superconductors in the **mixed state** between H_{c1} and H_{c2} . Abrikosov's prediction of the flux lattice was confirmed experimentally, showing not just that vortices occur, but they also tend to align in a regular triangular lattice, as predicted by the theory. This periodic lattice of vortices was perhaps the first example in physics of *emergent phenomena* in *complex systems*; the fact that sufficiently complex systems exhibit

a variety of novel phenomena on different length scales. These phenomena effectively arise from *self-organization* on the macroscopic length scale.

In type II superconductors the thermodynamic phase transition at H_{c2} is second order (see problem 4.1). Therefore we can expect that the Ginzburg-Landau order parameter ψ is small in magnitude just below H_{c2} and reaches zero exactly at H_{c2} .⁷ Therefore, the magnetization M will also be small at a magnetic field just below H_{c2} (since ψ is near zero, the superfluid density n_s and the screening supercurrents will also vanish at H_{c2}), as can be seen in Fig. 1.8. Therefore to a good approximation we can assume that

$$\mathbf{B} = \mu_0 \mathbf{H}, \quad (2.77)$$

where $\mathbf{H}$ is, as usual, the applied field given by the external apparatus as in Fig. 1.6. This also implies that sufficiently near to H_{c2} we can neglect any spatial variations in the B-field, $\mathbf{B}(\mathbf{r})$ and just treat it as a constant,

$$\mathbf{B} = (0, 0, B). \quad (2.78)$$

It will be convenient to express the corresponding vector potential $\mathbf{A}$ in the **Landau gauge** as

$$\mathbf{A}(\mathbf{r}) = (0, xB, 0). \quad (2.79)$$

In which case the Ginzburg-Landau equation, Eq. ?? becomes

$$-\frac{\hbar^2}{2m^*} \left(\nabla + \frac{2eBi}{\hbar} x \mathbf{j} \right) \cdot \left(\nabla + \frac{2eBi}{\hbar} x \mathbf{j} \right) \psi(\mathbf{r}) + a(T)\psi + b|\psi|^2\psi = 0, \quad (2.80)$$

where, as usual, $\mathbf{j}$ is the unit vector in the y direction.

Now if we are infinitesimally below H_{c2} , then ψ is essentially zero and we can drop the cubic term, $b|\psi|^2\psi$. All the other terms are linear in ψ and so we have **linearized** the equation. Expanding out the bracket (paying attention to the commutation of ∇ and $x\mathbf{j}$) gives

$$-\frac{\hbar^2}{2m^*} \left(\nabla^2 + \frac{4eBi}{\hbar} x \frac{\partial}{\partial y} - \frac{(2eB)^2}{\hbar^2} x^2 \right) \psi(\mathbf{r}) + a(T)\psi = 0. \quad (2.81)$$

Introducing the cyclotron frequency,

$$\omega_c = \frac{2eB}{m^*}, \quad (2.82)$$

and noting that $a(T)$ is negative since we are at a temperature below the zero field T_c , the equation can be written in the form,

$$\left(-\frac{\hbar^2}{2m^*} \nabla^2 - \hbar\omega_c i x \frac{\partial}{\partial y} + \frac{m^*\omega_c^2}{2} x^2 \right) \psi(\mathbf{r}) = |a|\psi(\mathbf{r}). \quad (2.83)$$

where $\xi(T)$ is the Ginzburg-Landau coherence length.

⁷In contrast this would not be the case for a type I superconductor, where the transition at H_c is first order. In these superconductors ψ jumps discontinuously from zero to a finite value at H_c .

Now Eq. 2.83 has the form of an eigenvalue equation, and is well known in quantum mechanics. It is equivalent to the Schrödinger equation for the wave function of a charged particle in a magnetic field, which has well known **Landau level** solutions (Ziman 1979). The solution has the form,

$$\psi(\mathbf{r}) = e^{i(k_y y + k_z z)} f(x), \quad (2.84)$$

which is a combination of plane waves in the y and z directions, combined with an unknown function of x , $f(x)$.

To find an equation for this function, $f(x)$ we substitute the trial solution into Eq. 2.83. We find that $f(x)$ obeys

$$-\frac{\hbar^2}{2m^*} \frac{d^2 f}{dx^2} + \left(\hbar \omega_c k_y x + \frac{m^* \omega_c^2}{2} x^2 \right) f = (|a| - \frac{\hbar^2 (k_y^2 + k_z^2)}{2m^*}) f. \quad (2.85)$$

The term in brackets on the left hand side can be rearranged by “completing the square”,

$$\left(\hbar \omega_c k_y x + \frac{m^* \omega_c^2}{2} x^2 \right) = \frac{m^* \omega_c^2}{2} (x - x_0)^2 - \frac{m^* \omega_c^2}{2} x_0^2 \quad (2.86)$$

where

$$x_0 = -\frac{\hbar k_y}{m \omega_c}. \quad (2.87)$$

Finally, moving all the constants over to the right hand side we find,

$$-\frac{\hbar^2}{2m^*} \frac{d^2 f}{dx^2} + \frac{m^* \omega_c^2}{2} (x - x_0)^2 f = (|a| - \frac{\hbar^2 k_z^2}{2m^*}) f. \quad (2.88)$$

Eq. 2.88 is just the Schrödinger equation for a simple Harmonic oscillator, except that the origin of coordinates is shifted from $x = 0$ to $x = x_0$. Therefore the term in square brackets on the right is just the energy of the oscillator,

$$\left(n + \frac{1}{2} \right) \hbar \omega_c = |a| - \frac{\hbar^2 k_z^2}{2m^*}, \quad (2.89)$$

or

$$\left(n + \frac{1}{2} \right) \hbar \omega_c + \frac{\hbar^2 k_z^2}{2m^*} = \dot{a}(T_c - T). \quad (2.90)$$

The corresponding functions $f(x)$ are just the wave functions of a simple Harmonic oscillator for each n , shifted by x_0 .

Imagine that we gradually cool a superconductor in an external field, H . At the zero field transition temperature, T_c , it will be impossible to satisfy Eq. 2.90 because of the zero point energy term $\hbar \omega_c/2$ on the right

hand side. A solution will only be possible when the temperature is far enough below T_c to achieve,

$$\frac{1}{2}\hbar\omega_c = \dot{a}(T_c - T), \quad (2.91)$$

corresponding to the lowest possible energy solution ($n = 0$, $k_z = 0$). This equation determines the depression in transition temperature in the magnetic field,

$$\begin{aligned} T_c(H) &= T_c(0) - \frac{1}{2\dot{a}}\hbar\omega_c \\ &= T_c(0) - \frac{2e\hbar\mu_0}{2\dot{a}m^*}H. \end{aligned} \quad (2.92)$$

Alternatively, we can start in a large external field, H , above H_{c2} which we gradually decrease (keeping the temperature fixed) until

$$\frac{1}{2}\hbar\frac{2eB}{m^*} = \dot{a}(T_c - T). \quad (2.93)$$

Therefore, rearranging,

$$\begin{aligned} \mu_0 H_{c2} = B_{c2} &= \frac{2m^*\dot{a}(T_c - T)}{\hbar^2} \frac{\hbar}{2e} \\ &= \frac{\Phi_0}{2\pi\xi(T)^2}. \end{aligned} \quad (2.94)$$

It is interesting to note that this result implies that at H_{c2} there is exactly one flux quantum (i.e. one vortex line), in each unit area $2\pi\xi(T)^2$. This expression also provides the simplest way to measure the Ginzburg-Landau coherence length $\xi(0)$ experimentally. Since $\xi(T) = \xi(0)t^{-1/2}$ where $t = |T - T_c|/T_c$,

$$\mu_0 H_{c2} = \frac{\Phi_0}{2\pi\xi(0)^2} \frac{T_c - T}{T_c} \quad (2.95)$$

and so by measuring the gradient of $H_{c2}(T)$ near to T_c one can easily find the corresponding $\xi(0)$.

It is also interesting to compare this expression for H_{c2} with the corresponding result for H_c which we found earlier. Using the Ginzburg-Landau expression for the total energy in the Meissner state we had found that

$$\begin{aligned} H_c &= \frac{\dot{a}}{(\mu_0 b)^{1/2}}(T_c - T) \\ &= \frac{\Phi_0}{2\pi\mu_0\sqrt{2}\xi\lambda} \end{aligned}$$

$$= \frac{H_{c2}}{\sqrt{2}\kappa}, \quad (2.96)$$

and therefore

$$H_{c2} = \sqrt{2}\kappa H_c \quad (2.97)$$

From this we can deduce that for superconductors with $\kappa > 1/\sqrt{2}$ we will have $H_{c2} > H_c$, and the phase transition will be second order with the order parameter growing continuously from zero at fields just below H_{c2} . I.e. we will have a type II superconductor. On the other hand, for superconductors with $\kappa < 1/\sqrt{2}$ we will have $H_{c2} < H_c$, and the phase transition will be first order phase transition at the field H_c , below which the order parameter jumps discontinuously to a finite value. The Abrikosov theory therefore immediately describes the difference between type I and type II superconductors,

$$\kappa \begin{cases} < \frac{1}{\sqrt{2}} & \text{type I} \\ > \frac{1}{\sqrt{2}} & \text{type II} \end{cases}$$

The linearized Ginzburg-Landau equation allows us to find at H_{c2} , but does not immediately tell us anything about the form of the solution below this field. To do this we must solve the non-linear equation, Eq. 2.80. This is very hard to do in general, but Abrikosov made a brilliant guess and came up with essentially the exact solution! He could see from the solutions to the linearized equation Eq. 2.80 that only the harmonic oscillator ground state solutions $n = 0$ and $k_z = 0$ will be significant. However there are still an infinite number of degenerate states, corresponding to the different possible k_y values,

$$\psi(\mathbf{r}) = C e^{i(k_y y)} e^{-(x-x_0)^2/\xi(T)^2}, \quad (2.98)$$

where C is a normalization constant. Here we have used the fact that the ground state wave function of a quantum harmonic oscillator is a gaussian function. The width of the gaussian solution to Eq. 2.88 turns out to be the GL coherence length $\xi(T)$.

Abrikosov's trial solution was to assume that we can combine these solutions into a **periodic** lattice. If we look for a solution which is periodic in y with period l_y , then we can restrict the values of k_y to

$$k_y = \frac{2\pi}{l_y} n \quad (2.99)$$

with n any positive or negative integer. The corresponding Landau level x shift is

$$x_0 = -\frac{2\pi\hbar}{m\omega_c l_y} n = -\frac{\Phi_0}{B l_y} n. \quad (2.100)$$

Therefore we can try a periodic solution

$$\psi(\mathbf{r}) = \sum_{n=-\infty, \infty} C_n e^{i(2\pi n y / l_y)} e^{-(x+n\Phi_0 / Bl_y)^2 / \xi(T)^2}. \quad (2.101)$$

In this solution we can view the parameters C_n as variational parameters which are to be chosen to minimize the total GL free energy of the system.

The solution above is periodic in y , but not necessarily periodic in x . Abrikosov noted that it can be made periodic in x provided the coefficients obey

$$C_{n+\nu} = C_n \quad (2.102)$$

for some integer ν . The period is l_x , where

$$l_x = \nu \frac{\Phi_0}{Bl_y}. \quad (2.103)$$

Abrikosov studied the simplest case, $\nu = 1$ which corresponds to a simple square lattice. Later it was shown that a slightly lower total energy is obtained for the $\nu = 2$ case and the minimum energy state corresponds to a simple triangular lattice. In each case the order parameter $\psi(\mathbf{r})$ goes to zero at one point in each unit cell, and there is exactly one flux quantum Φ_0 per unit cell. Therefore the solution is a **periodic lattice of vortices**. This is illustrated in Fig. 2.7.

Experimental evidence for the Abrikosov flux lattice comes from a variety of methods. In a “flux decoration” experiment small paramagnetic particles are dusted onto the surface of the superconductor (just like the children’s experiment to see magnetic fields of a bar magnet using iron filings on paper above the magnet!). The particles concentrate into the points of highest magnetic field, i.e. the vortices. Other similar methods involve scanning a small SQUID loop or Hall probe just above the surface of the superconductor to directly measure the variation of the local flux density $\mathbf{B}(\mathbf{r})$. For most ordinary type II low T_c superconductors, such as Pb or Nb, these experiments indeed show that the vortices form a fairly regular hexagonal lattice. The lattice can be periodic over quite long length scales, but is disrupted now and then by defects. These defects (exactly analogous to crystal dislocations or point defects) tend to concentrate near defects in the underlying crystal lattice (such as grain boundaries, impurities etc.).

Another method of observing the order in the flux lattice is to use neutron scattering. The neutrons have a magnetic moment, and so are sensitive to the magnetic field $\mathbf{B}(\mathbf{r})$. If this is periodic, as in the Abrikosov flux lattice, then there will be diffraction. The diffraction pattern can be used to find the geometry of the flux lattice. Again the majority of systems studied are found to have triangular lattices. But, interestingly, a square lattice has now been found in a few recently discovered superconductors. These are: the ‘borocarbide’ system $\text{ErNi}_2\text{B}_2\text{C}$, the high temperature superconductor $\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$ and the possible p-wave strontium ruthenate Sr_2RuO_4 . All of these have been found to have square vortex lattices in at least some range of external fields. It may be that this is simply because

of small corrections to the original Abrikosov theory (such as terms omitted in the standard GL equations, such as higher powers of ψ , like $|\psi|^6$, or higher order gradients). In fact the square lattice and triangular lattice have energies differing by less than 1% in the Abrikosov solution. But in some cases it seems more likely to be due to the fact that the underlying form of superconductivity is “unconventional”, meaning that the Cooper pairs have a different symmetry from the normal BCS case. We shall introduce these ideas briefly in chapter 7.

Finally, we should note that while the Abrikosov solution is essentially exact just below H_{c2} , it cannot necessarily be applied far away from there, such as at H_{c1} . As we have seen, near to H_{c2} the vortices are close together, separated by distances of order the coherence length $\xi(T)$. Effectively they are so densely packed that their cores are essentially touching. On the other hand, just at H_{c1} we have very few vortices in the entire sample, and so they are well separated. We can estimate of the lower critical field H_{c1} from the energy balance for the very first few vortices to enter a superconductor in the Meissner phase. One can show that a single London vortex has an energy of approximately (see exercise 3.3)

$$E = \frac{\Phi_0^2}{4\pi\mu_0\lambda} \ln\left(\frac{\lambda}{\xi}\right) \quad (2.104)$$

per unit length. Therefore in a superconductor with N/A flux lines per unit area and thickness L , there will be a total energy cost EN/A per unit volume due to the vortices. But on the other hand, each vortex carries a flux Φ_0 , and so the average magnetic induction in the sample is $B = \Phi_0 N/A$. The magnetic work gained by the presence of the vortices is $\mu_0 H dM = H dB$ (at constant H). Energy balance therefore favours the presence of the vortices when

$$E \frac{N}{A} < H \Phi_0 \frac{N}{A}. \quad (2.105)$$

Thus it becomes energetically favourable for the vortices to enter the sample when $H > H_{c1}$, where

$$H_{c1} = \frac{\Phi_0}{4\pi\mu_0\lambda^2} \ln\left(\frac{\lambda}{\xi}\right). \quad (2.106)$$

This is obviously the lower critical field, and can be simply expressed as

$$H_{c1} = \frac{H_c}{\sqrt{2}\kappa} \ln(\kappa). \quad (2.107)$$

This expression is only valid when $\kappa \gg 1/\sqrt{2}$, i.e. in the London vortex limit.

2.10 Thermal Fluctuations

The Ginzburg-Landau theory as described above is purely a **mean-field theory**. It neglects **thermal fluctuations**. In this it is therefore similar to the

Curie-Weiss or Stoner models in the theory of magnetism (Blundell 2001). But in fact it is a great strength of the GL theory that it can easily be extended so that these fluctuations can be included. It is much simpler to include these in the GL theory than in the more complex BCS theory.

In the mean-field approach, we must always find the order parameter $\psi(\mathbf{r})$ which minimizes the total free energy of the system. As discussed above, this is a functional minimization. The total free energy of the system $F[\psi]$, in Eq. 2.29, is a functional of the complex order parameter, $\psi(\mathbf{r})$, meaning that it depends on an infinite number of variables: the values of ψ all possible points $\mathbf{r}$. As we saw, the condition for minimizing the free energy is that the functional derivatives given in Eq. 2.37 are zero.

To go beyond this mean-field approach we must include fluctuations of $\psi(\mathbf{r})$ close to this minimum. For example if we make a small variation in $\psi(\mathbf{r})$, such as $\psi(\mathbf{r}) \rightarrow \psi'(\mathbf{r}) = \psi(\mathbf{r}) + \delta\psi(\mathbf{r})$, then we expect that the energy of the system represented by $\psi'(\mathbf{r})$ would be very similar to that represented by $\psi(\mathbf{r})$. If the total energy difference is small, or no more than $k_B T$, then we might expect that in thermal equilibrium the system would have some probability to be in state $\psi'(\mathbf{r})$. We need to define an effective probability for each possible state. Clearly this must be based on the usual Boltzmann probability distribution and so we expect that,

$$P[\psi] = \frac{1}{Z} e^{-\beta F[\psi]} \quad (2.108)$$

is the probability density for the system to have order parameter $\psi(\mathbf{r})$. It is again a functional of $\psi(\mathbf{r})$ as indicated by the square brackets.

The partition function Z is the normalization factor in this expression. Formally it is a **functional integral**,

$$Z = \int \mathcal{D}[\psi] \mathcal{D}[\psi^*] e^{-\beta F[\psi]}. \quad (2.109)$$

We can treat the integrals over ψ and ψ^* as formally separate for the same reason that we could view functional derivatives with respect to ψ and ψ^* as formally independent. It is really allowed because in fact we have to specify two independent real functions, the real and imaginary parts of ψ at each point $\mathbf{r}$: $\text{Re}[\psi(\mathbf{r})]$ and $\text{Im}[\psi(\mathbf{r})]$.

What is the meaning of the new integration symbols, like $\mathcal{D}[\psi]$ in Eq. 2.109? We are technically integrating over an infinite number of variables, the values of $\psi(\mathbf{r})$ at every point $\mathbf{r}$. It is difficult (and beyond the scope of this book!) to make this idea mathematically rigorous. But we can find an intuitive idea of what this means by supposing that we only had a discrete set of points in space, $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N$. We could define values of ψ and ψ^* at each point, and then calculate a Boltzmann probability. The approximate partition function for this discrete set would be the multiple integral

$$Z(N) = \int d\psi(\mathbf{r}_1) d\psi(\mathbf{r}_1^*) \int d\psi(\mathbf{r}_2) d\psi(\mathbf{r}_2^*) \dots \int d\psi(\mathbf{r}_N) d\psi(\mathbf{r}_N^*) e^{-\beta F[\psi]}. \quad (2.110)$$

The full functional integral is a limit in which we make the set of points infinitely dense (in fact even possibly uncountably infinite!), defining

$$Z = \lim_{N \rightarrow \infty} Z(N). \quad (2.111)$$

One way that this infinite product of integrals might be accomplished is through the Fourier transforms of $\psi(\mathbf{r})$ and $\psi^*(\mathbf{r})$. If we define $\psi_{\mathbf{k}}$ by

$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} \psi_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} \quad (2.112)$$

then specifying the parameters $\psi_{\mathbf{k}}$ and $\psi_{\mathbf{k}}^*$ at every wave vector $\mathbf{k} = (2\pi n_x/L_x, 2\pi n_y/L_y, 2\pi n_z/L_z)$ (or equivalently the real and imaginary parts), defines the full functions $\psi(\mathbf{r})$ and $\psi^*(\mathbf{r})$. In this representation we can write the partition function as

$$Z = \prod_{\mathbf{k}} \left(\int d\psi_{\mathbf{k}} d\psi_{\mathbf{k}}^* \right) e^{-\beta F[\psi]}. \quad (2.113)$$

Again there are an infinite number of integrals, two for each point $\mathbf{k}$.

As an example of the sort of thermal fluctuation effects that can be calculated within this formalism we shall consider the specific heat of a superconductor near to T_c . For a superconductor in zero magnetic field we have the free energy functional,

$$F[\psi] = \int d^3r \left(\frac{\hbar^2}{2m^*} |\nabla \psi|^2 + a|\psi|^2 + \frac{b}{2} |\psi|^4 \right) \quad (2.114)$$

(dropping the constant normal state free energy F_n , which will be irrelevant here). Writing this in terms of the Fourier coefficients ψ_{bfk} we find

$$F[\psi] = \sum_{bfk} \left(\frac{\hbar^2 k^2}{2m^*} + a \right) \psi_{\mathbf{k}}^* \psi_{\mathbf{k}} + \frac{b}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3} \psi_{\mathbf{k}_1}^* \psi_{\mathbf{k}_2}^* \psi_{\mathbf{k}_3} \psi_{\mathbf{k}_1 + \mathbf{k}_2 - \mathbf{k}_3}, \quad (2.115)$$

which could in principle be inserted directly into Eq. 2.113. In general this would be very difficult, and requires either massive numerical Monte Carlo simulation, or some other approximation. The simplest approximation that we can make is to make the **gaussian approximation**, in which we neglect the quartic (b) term in the free energy. In this approximation we find a simple result

$$Z = \prod_{\mathbf{k}} \int d\psi_{\mathbf{k}} d\psi_{\mathbf{k}}^* \exp \left\{ -\beta \left(\frac{\hbar^2 k^2}{2m^*} + a \right) \psi_{\mathbf{k}}^* \psi_{\mathbf{k}} \right\}. \quad (2.116)$$

Changing variables to the two real functions, $\text{Re}[\psi_{\mathbf{k}}]$ and $\text{Im}[\psi_{\mathbf{k}}]$ gives

$$Z = \prod_{\mathbf{k}} \int d\text{Re}[\psi_{\mathbf{k}}] d\text{Im}[\psi_{\mathbf{k}}] \exp \left\{ -\beta \left(\frac{\hbar^2 k^2}{2m^*} + a \right) (\text{Re}[\psi_{\mathbf{k}}]^2 + \text{Im}[\psi_{\mathbf{k}}]^2) \right\}, \quad (2.117)$$

and so for each $\mathbf{k}$ the integral is just a two dimensional gaussian integral. These can be done exactly, resulting in the partition function

$$Z = \prod_{\mathbf{k}} \frac{\pi}{\beta \left(\frac{\hbar^2 k^2}{2m^*} + a \right)}. \quad (2.118)$$

From the partition function it is possible to calculate all thermodynamic quantities of interest. For example, the total internal energy is given by Eq. ??,

$$\begin{aligned} U &= -\frac{\partial \ln Z}{\partial \beta} \\ &= +k_B T^2 \frac{\partial \ln Z}{\partial T} \\ &\sim -\sum_{\mathbf{k}} \frac{1}{\left(\frac{\hbar^2 k^2}{2m^*} + a \right)} \frac{da}{dT}, \end{aligned} \quad (2.119)$$

where in the last step we have kept only the most important contribution which comes from the change of the Ginzburg Landau parameter a with T , $da/dT = \dot{a}$.

The gaussian approximation for the heat capacity near to T_c is found by differentiating again, giving

$$\begin{aligned} C_V = \frac{dU}{dT} &= \sum_{\mathbf{k}} \frac{1}{\left(\frac{\hbar^2 k^2}{2m^*} + a \right)^2} \dot{a}^2. \\ &= \frac{V}{(2\pi^3)} \frac{\dot{a}^2}{a^2} \int d^3k \frac{1}{(1 + \xi(T)^2 k^2)^2} \\ &\sim \frac{V}{(2\pi^3)} \frac{\dot{a}^2}{a^2} \frac{1}{\xi(T)^3} \\ &\sim \frac{1}{(T - T_c)^2} |T_c - T|^{3/2} \\ &\sim \frac{1}{|T - T_c|^{1/2}}, \end{aligned} \quad (2.120)$$

(where, for simplicity, we have ignored the numerical multiplying prefactors). This shows that the thermal fluctuations can make a very large contribution to the heat capacity, essentially diverging at the critical temperature T_c . If we sketch this behaviour we see that the thermal fluctuations

make a large difference to the original mean-field specific heat of Fig. 2.4. As can be seen in Fig. ???. In fact, once the fluctuations are included the the heat capacity of superconductors becomes much more similar to the heat capacity of superfluid ^4He at T_c , as shown in Fig. ???.⁸

Experimentally these thermal fluctuations are very difficult to see in standard “low T_c ” superconductors, such as Pb or Nb. It is possible to estimate the range of temperatures near to T_c where these fluctuations are significant. This temperature range, T_G is known as the Ginzburg critereon. In 1960 Ginzburg found that this temperature range is extremely small, i.e. much less than $1\mu\text{K}$ for most low T_c superconductors. Therefore we can say that the original mean-field approach to the Ginzburg Landau equations is perfectly well justified. However, in the high temperature superconductors, discovered in 1986, the coherence length $\xi(0)$ is very small (Table 2.1), of order just a few Angstroms. It turns out that the corresponding Ginzburg temperature range, T_G , is of order $1 - 2\text{K}$. Therefore is is quite possible to see such thermal fluctuation effects in these systems. The specific heat near T_c clearly shows thermal critical fluctuations, as shown in Fig. 2.9. In fact in these experimental results very good agreement is found using the value of the critical exponent α given by the three dimensional XY model predictions, exactly as is the case in superfluid ^4He , Fig. ???. The gaussian model exponent $\alpha = 1/2$ does not fit at all as well. Another example of thermal fluctuation effects can be seen in the resistivity, $\rho(T)$, just above T_c . Thermal fluctuations make $\rho(T)$ begin to bend down towards zero even at temperatures quite far above T_c . This downward bending is clearly visible in the resistivity curve of the $T_c = 135\text{K}$ superconductor $\text{HgBa}_2\text{Ca}_2\text{Cu}_3\text{O}_{8+\delta}$ shown in Fig. 1.2.

2.11 Vortex Matter

Another very important consequence of thermal fluctuations occurs in the mixed state of high temperature superconductors. As we have seen, Abrikosov’s flux lattice theory shows that the vortices align in a periodic lattice arrangement, essentially like a crystal lattice, either triangular or square. However, this is again a mean field approximation! We must, in principle, again include the effects of thermal fluctuations.

The theories of the resulting **vortex matter** states show a very wide range of possibilities. It is still possible to talk about the vortices, but now they themselves form a variety of different states, including liquid and glassy (random, but frozen) states, as well as nearly perfectly ordered

⁸In fact, the gaussian theory, as outlined above, is not truly correct since it dropped the $|\psi|^4$ terms in the Ginzburg-Landau free energy. When these terms are included the resulting theory is known in statistical physics as the XY or $O(2)$ model. Its true critical behaviour near T_c can be calculated with various methods based on the renormalization group. The resulting critical exponent for specific heat α is very small and very different from the $\alpha = 1/2$ which is given by the gaussian approximation of Eq. 2.120.

crystalline states. It is believed that the flux lattice never has true crystalline order, and thermal fluctuations always lead to an eventual loss of long ranged order in the periodic structure (although in practice periodicity can be quite well extended). A full discussion of these topics requires a whole book in itself (Singer and Schneider 2000), and there also are many extensive review papers (Blatter 1994).

Unfortunately these thermal fluctuations have been disastrous for commercial applications of high T_c superconductors in high current wires and electromagnets (Yeshurun 1998).⁹ The problem is that thermal fluctuations lead to motion of the vortices, and this leads to a source of energy dissipation. Therefore the resistivity is not zero for high T_c superconductors in a magnetic field. The problem also occurs in low T_c superconductors, but to a much lesser extent. In these systems the energy dissipation due to motion of vortices can be reduced or eliminated by providing **pinning centres** which “pin” the vortex lattice and prevent it from moving. Typically these are just impurities, or naturally occurring crystal defects such as grain boundaries and dislocations.

To see why motion of vortices leads to energy dissipation is necessary to see that a current density $\mathbf{j}$ flowing through the vortex lattice (perpendicular to the magnetic field) leads to a Lorentz (or Magnus) force on each vortex. The overall force is

$$\mathbf{f} = \mathbf{j} \times \hat{\mathbf{B}} \quad (2.121)$$

per unit volume of the vortex lattice. This will tend to make the vortex liquid flow in the direction perpendicular to the current as shown below on the left, as illustrated in Fig. ??.

Unfortunately if the vortices flow in response to this force, work is done and there is energy dissipation. To calculate the work, consider a loop of superconducting wire, with a current flowing around the wire. Vortices will tend to drift transversely across the wire, say entering on the inner side of the wire and drifting over to the outer side. This is illustrated in Fig. ??. Each vortex carries a magnetic flux Φ_o , and so the total magnetic flux in the ring Φ changes by Φ_o with each vortex that crosses from one side of the wire to the other. But by elementary electromagnetism there is an EMF induced in the wire given by $\mathcal{E} = -d\Phi/dt$. Power is dissipated, at a rate given by $P = \mathcal{E}I$ where I is the total current. Therefore vortex motion directly leads to **finite resistance**! In the mixed state, superconductors only have truly zero resistance when the vortices are pinned and unable to move.

⁹Perhaps this is not the only difficulty with commercial applications of high T_c superconductivity. The materials are brittle and cannot easily be made into wires. Nevertheless these problems have been gradually overcome, and now high T_c superconducting wires are beginning to make a serious entry into commercial applications. For example, at least one US city receives part of its electrical power through underground superconducting cables. Some microwave receivers, such as in some masts for mobile phones beside motorways, also use superconducting devices operating at liquid nitrogen temperatures.

In high T_c superconductors the thermal motion of vortices leads to especially bad pinning and hence a significant resistivity in the mixed state. To make matters worse, the lower critical field H_{c1} is tiny, often less than the Earth's magnetic field, and so vortices can never be truly eliminated. At high temperatures and near to H_{c2} it is believed that the vortex matter is in a liquid state, and so the vortices can move freely and pinning is essentially impossible.¹⁰ Lowering the temperature, or going further away from H_{c2} the vortex matter appears to “freeze” into a glassy state. A glass is random spatially, but frozen in time. Since glass is effectively rigid the vortices cannot move and pinning is able to largely prevent flux motion. Therefore in this state the resistivity is quite low. Unfortunately, even in this glassy vortex state the resistivity is not fully zero, since **flux creep** can occur. The random pinning force provides a set of energy barriers to vortex motion, but thermal motions mean that from time to time the vortices can hop over the local energy barrier and find a new configuration.¹¹ The line in the (H, T) phase diagram where the glassy phase occurs is called the “irreversibility line”, as shown in Fig. 2.10. Something approaching zero resistivity is approached only well below this line. This effectively limits the useful magnetic fields for applications of high T_c superconductivity, to very much less than the hundreds of Tesla that one might have expected from the nominal values of $\mu_0 H_{c2} > 100T$, such as one might expect from Table 2.1.

2.12 Summary

We have seen how the Ginzburg Landau theory provides a mathematically rather simple picture with which to describe quite complex phenomena in superconductivity. In terms of the phenomenological order parameter, $\psi(\mathbf{r})$, and four empirically determined parameters (a, b, m^* and T_c) we can construct a full theory of superconductivity which encompasses fully phenomena such as the Abrikosov flux lattice, flux quantization and from which one can “derive” the London equation.

The power of the theory is also apparent in the way it can be modified to incorporate thermal fluctuations, including critical phenomena and vortex matter physics. It should be noted that these areas are still highly active areas of experimental and theoretical activity. Even some very simple and fundamental questions are still hotly debated, such as the various vortex phases occurring in high temperature superconductors. These also have important implications for potential commercial applications of these materials.

¹⁰A liquid can always flow around any impurities and so pinning centres have no effect in the vortex liquid state.

¹¹The process is presumably analogous to the way that window glass in medieval cathedrals appears to have gradually flowed downwards over timescales of hundreds of years.

2.13 Further Reading

Magnetic work in thermodynamics is discussed in detail in the textbooks by Mandl (1987) and Callen (1960).

The idea of order parameters and Ginzburg Landau theory in general, including superconductors, are discussed in Chakin and Lubensky (1995) and Anderson (1984). Applications of Ginzburg Landau theory to vortex states and other problems are covered in detail by de Gennes (1966). Other books are also useful, such as Tilley and Tilley (1990), Tinkham (1996). In fact almost all textbooks on superconductivity have at least one chapter dealing with Ginzburg Landau theory and its predictions.

Thermal fluctuations and critical phenomena are in themselves huge fields of study. A good introductory course is Goldenfeld (1992), while the books Amit (1984) and Ma (1976) are very comprehensive. These books discuss very general classes of theoretical models, but the Ginzburg Landau theory we have discussed is equivalent to the model they call XY or $O(2)$.

For a modern view of thermal fluctuation phenomena and the problems of vortex matter physics, especially in its application to high temperature superconductors, see the book by Singer and Schneider (2000), or the review articles by Blatter (1994) and Yeshurun (1996).

2.14 Exercises

(4.1) (a) For a type I superconductor $H_c(T)$ is the boundary between normal metal and superconductor in the H, T phase diagram. Everywhere on this boundary thermal equilibrium requires

$$G_s(T, H) = G_n(T, H).$$

Apply this equation and $dG = -SdT - \mu_0 M dH$ to two points on the $H - T$ phase boundary (T, H) and $(T + \delta T, H + \delta H)$, as illustrated in Fig. 2.12. Hence show that:

$$-S_s \delta T - \mu_0 M_s \delta H = -S_n \delta T - \mu_0 M_n \delta H$$

when δT and δH are small, and where $S_{s/n}$ $M_{s/n}$ are the superconducting and normal state entropy and magnetization respectively.

(b) Using part (a), and $M_n = 0$, and $M_s = -H$ show that the latent heat per unit volume for the phase transition, $L = T(S_n - S_s)$, is given by

$$L = -\mu_0 T H_c \frac{dH_c(T)}{dT}$$

where the phase boundary curve is $H_c(T)$. (This is exactly analogous to the Clausius-Claperyon equation in a gas-liquid phase change except H replaces P and $-\mu_0 M$ replaces V . See Mandl (1987) p 228).

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(4.2) Find $|\psi|^2$, the free energy $F_s - F_n$, the entropy and the heat capacity of a superconductor near T_c , using the bulk Ginzburg-Landau free energy. Sketch their variations with temperature assuming that $a = \dot{a} \times (T - T_c)$ and $\dot{a}$ and b are constant near T_c .

(4.3) (a) Show that for one-dimensional problems, such as the surface discussed in sec. 4.5, the Ginzburg Landau equations for $\psi(x)$ can be rewritten as:

$$-\frac{d^2}{dy^2}f(y) - f(y) + f(y)^3 = 0$$

where $x = y\xi(T)$, $f(y) = \psi(y\xi)/\psi_0$, and $\psi_0 = \sqrt{|a|/b}$.

(b) Verify that

$$f(y) = \tanh(y/\sqrt{2})$$

is a solution to the equation in (b) corresponding to the boundary condition $\psi(0) = 0$. Hence sketch $\psi(x)$ near the surface of a superconductor.

(c) Often the surface boundary condition of a superconductor is not $\psi(x) = 0$, but $\psi(x) = C$ where C is a numerical constant. Show that if $C < \psi_0$ we can just translate the solution from Prob. 4.2 sideways to find a valid solution for any value of C in the range $0 \leq C < \psi_0$.

(d) In the proximity effect a metal (in the half-space $x > 0$) is in contact with a superconductor (occupying the region $x < 0$). Assuming that the normal metal can be described by a Ginzburg-Landau model but with $a > 0$, show that the order parameter $\psi(x)$ induced in the metal by the contact with the superconductor is approximately

$$\psi(x) = \psi(0)e^{-x/\xi(T)}$$

where $\hbar^2/2m^*\xi(T)^2 = a > 0$, and $\psi(0)$ is the order parameter of the superconductor at the interface.

(4.4) (a) In Eq. 2.120 we showed that the gaussian model gives a divergence in specific heat of the form

$$C_V \sim \frac{1}{|T - T_c|^\alpha}$$

with $\alpha = 1/2$. Repeat the steps given in Eq. 2.120 for the case of a two dimensional system, and show that in this case the gaussian model predicts $\alpha = 1$.

(b) This argument can also be extended easily to the general case of d -dimensions. By replacing the $\mathbf{k}$ sum in Eq. 2.120 by an integral of the form

$$\sum_{\mathbf{k}} \rightarrow \frac{1}{(2\pi^d)} \int d^d k$$

show that in d dimensions we have the critical exponent

$$\alpha = 2 - \frac{d}{2}.$$

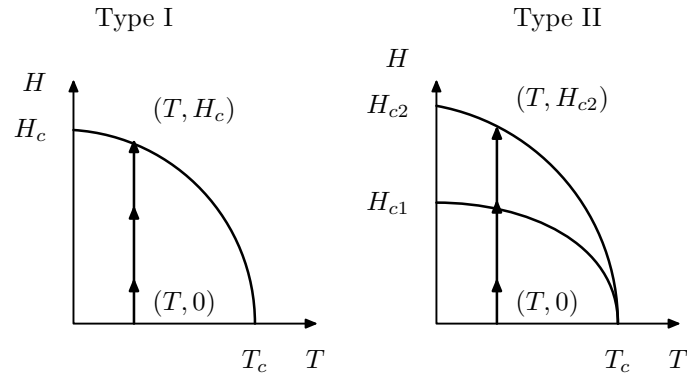


Fig. 2.1 We obtain the condensation energy for superconductors by thermodynamic integration of the Gibbs free energy along the contours in the (T, H) plane, as shown above.

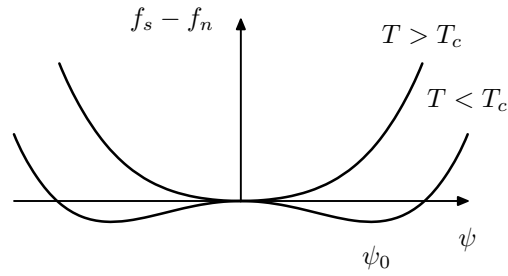


Fig. 2.2 Free energy difference between the normal and superconducting states (per unit volume) as a function of the order parameter ψ . For $T < T_c$ the free energy has a minimum at ψ_0 , while for $T > T_c$ the only minimum is at $\psi = 0$.

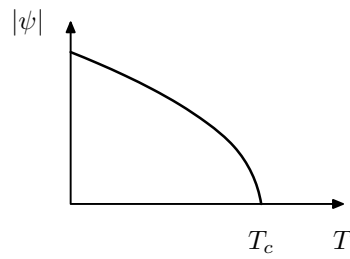


Fig. 2.3 Order parameter magnitude, $|\psi|$, as a function of temperature in the Ginzburg Landau model.

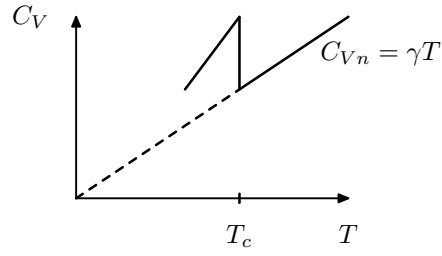


Fig. 2.4 Specific heat of a superconductor near T_c in the Ginzburg Landau model. Above T_c the specific heat is given by the Sommerfeld theory of metals, $C_{Vn} = \gamma T$. At T_c there is a discontinuity and a change of slope.

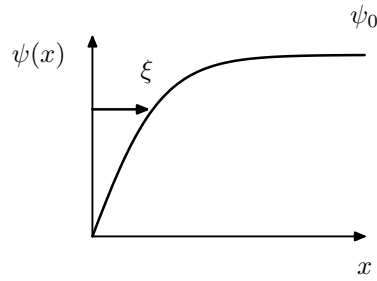


Fig. 2.5 Order parameter of a superconductor near a surface. It recovers to its bulk value ψ_0 over a length scale of the coherence length, ξ .

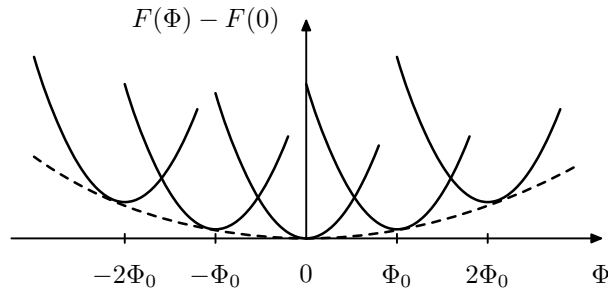


Fig. 2.6 Flux quantization in a superconducting ring. Metastable energy minima exist when the flux is an integer multiple of the flux quantum $\Phi_0 = h/2e$. There is an overall background increase with Φ corresponding to the self-inductance of the ring, making the zero flux state $\Phi = 0$ the global energy minimum. Thermal fluctuations and quantum tunnelling allow transitions between neighbouring meta-stable energy minima.

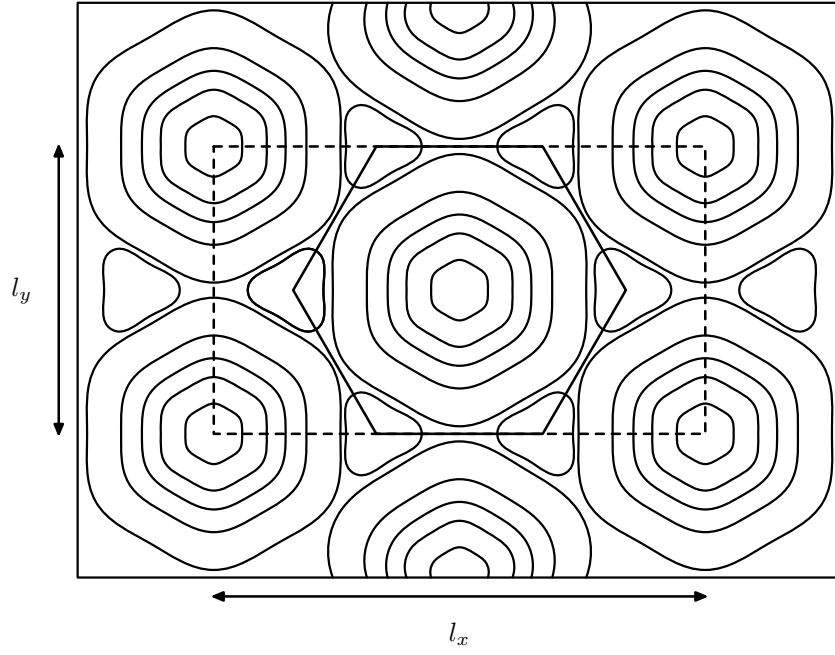


Fig. 2.7 The Abrikosov flux lattice. The figure shows the amplitude of the order parameter, $|\psi(\mathbf{r})|^2$ for the lowest energy triangular lattice. Each triangular unit cell contains one quantum, Φ_0 , of magnetic flux and contains one vortex where $\psi(\mathbf{r}) = 0$. In terms of the l_x and l_y lattice periodicities used in Sec. 4.9, $l_y = \sqrt{3}l_x$, and the rectangular unit cell $l_x \times l_y$ contains two vortices and two flux quanta.

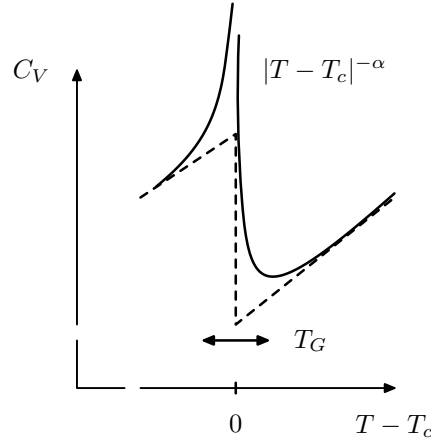


Fig. 2.8 Specific heat of a superconductor near to T_c in the gaussian approximation. The mean field Ginzburg-Landau theory gives a discontinuity at T_c . ut this is supplemented by a thermal fluctuation contribution which diverges like $|T - T_c|^{-\alpha}$ with $\alpha = 1/2$. The full renormalization group treatment (ignoring magnetic field terms) shows that α is given by the three-dimensional XY model value, exactly as in superfluid helium shown in Fig. ??.

Fig. 2.9 Experimental heat capacity of the high temperature superconductor $\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$ near to T_c . In zero magnetic field the experimental data fits very well the predictions of the three dimensional XY model. An external magnetic field (inset) removes the singularity, but does not visibly reduce T_c significantly. Reproduced from Overend, Howson and Lawrie (1994), with permission.

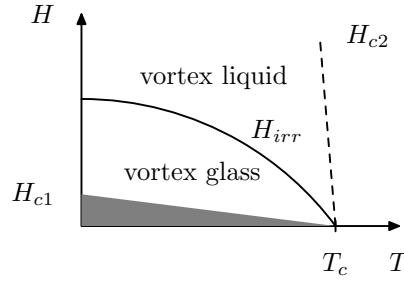


Fig. 2.10 Proposed magnetic phase diagram of some high T_c superconductors. Below H_{c2} vortices form, but are in a liquid state, leading to finite resistance of the superconductor. Below the ‘irreversibility line’ the vortices freeze (either into a glassy or quasi periodic flux lattice). In this state resistivity is still finite, due to flux creep, but becomes negligible far below the irreversibility line. H_{c1} is extremely small.

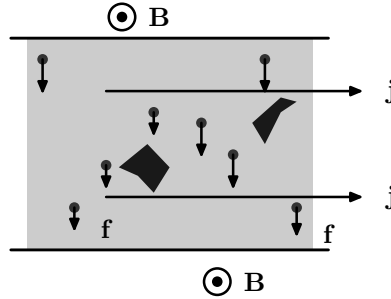


Fig. 2.11 Energy dissipation due to vortex flow in superconductors. Every vortex experiences a Lorentz (Magnus) force perpendicular to the supercurrent direction. This causes the vortices to drift sideways across the wire, unless pinned by defects. For each vortex which crosses the wire from one side to the other, a certain amount of work is done, and energy is dissipated.

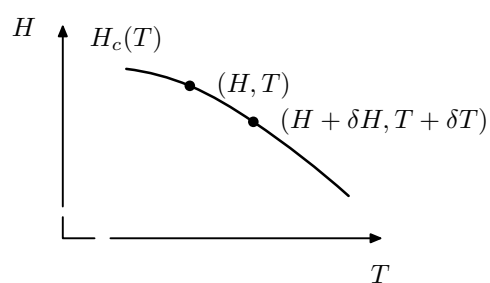


Fig. 2.12 Exercise 4.1. Consider the Gibbs free energy at the points shown on the normal-superconducting phase boundary of a type I superconductor, (T, H) and $T + \delta T, H + \delta H$. At both points equilibrium requires that both normal and superconducting Gibbs free energies must be equal: $G_n(T, H) = G_s(T, H)$.